

Aman Pandey Sanjeev Kumar

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Education

PURDUE UNIVERSITY
Master of Science in Electrical and Computer Engineering
Specialization: Artificial Intelligence & Machine Learning

West Lafayette, Indiana
May 2026

Master of Science in Industrial Engineering
Specialization: Operations Research

★ **Relevant Coursework:** Machine Learning Systems, Deep Learning, Reinforcement Learning, AI Hardware, Linear Programming, Linear Algebra, Advanced Probability

Technical Skills

Programming & GPU: Python, Pytorch, C++, CUDA C++, Triton, CuTe DSL, CUTLASS, cuBLAS/cuBLASLt
AI Agents & Serving: Claude Code, OpenCode, Codex, OpenRouter, LiteLLM, PyTorch, vLLM, SGLang, TensorRT-LLM, FlashInfer
Agentic Systems: tool calling, MCP servers, model routing, eval harnesses, observability, HITL approvals, RAG integrations
Systems & Deployment: Git, GitHub, Linux, Bash, Docker, Kubernetes, Slurm
Compilers & Profiling: MLIR, OpenXLA, TVM, Nsight Compute, Nsight Systems

Projects

GPU Mode B200 QR Decomposition: Long Horizon Agentic CUDA Engineering | [GitHub](#) | *Challenge*

- ★ Built a **multi-agent CUDA workflow** using OpenCode, Codex, LiteLLM routes, and coding/review agents.
- ★ Designed **permission gates, worktree handoffs**, runbooks, and reject logs for reproducible agent output.
- ★ Wrapped **Modal B200/Popcorn** actions with budget checks, source hashes, smoke tests, and safe artifacts.
- ★ Implemented **batched FP32 QR kernels** with compact-Householder $n=32/n=176$ paths and `torch.geqr` fallbacks.
- ★ Reduced local geomean by **23.7%** ($24.61 \rightarrow 18.79$ ms) while preserving reconstruction and orthogonality.
- ★ Validated triangularity, rank-deficient, clustered, mixed-batch, and seed-fuzz cases with repeatable gates.

Cross-DSL GPU Kernel Engineering for LLM Inference | [GitHub](#)

- ★ Implemented FP32 RMSNorm, decode GEMV/GEMM, and RoPE-KV kernels in **CUDA C++, Triton, and CuTe DSL**.
- ★ Built shared **PyTorch references, tests, CUDA-event benchmarks**, and **cuBLAS/cuBLASLt** decode baselines.
- ★ Profiled backend tradeoffs: CUDA RMSNorm neared **peak DRAM bandwidth**; Triton led mixed decode shapes.
- ★ Fused **RoPE, GQA page translation, and paged-KV writes in CUDA**, eliminating eager intermediates.

nano-vLLM++: Single-GPU LLM Inference Runtime | [GitHub](#)

- ★ Built a **single-GPU inference runtime** with request state machines, prefill/decode separation, and token-budget scheduling.
- ★ Implemented **KV-cache metadata management** with fixed-size block tables, reference counting, and chained-hash prefix detection.
- ★ Added **BF16/NF4/INT8 loading** and per-channel INT8/packed-INT4 prototypes; NF4 cut Qwen3-0.6B memory **1.11→0.53 GB**.
- ★ Exposed **OpenAI-compatible FastAPI endpoints** with **TTFT, TPOT, E2E**, queue-depth, and KV-block metrics.

FlashAttention-2 Triton Implementation | [GitHub](#)

- ★ Built a fused Triton FA2 kernel with causal masking and end-to-end **forward/backward passes** for 4K context.
- ★ Tiled QK^T and V to SRAM with online softmax, eliminating $O(N^2)$ HBM materialization bottlenecks.
- ★ Recomputed backward attention weights via saved **log-sum-exp**, avoiding $O(N^2)$ intermediate storage.
- ★ Profiled against PyTorch SDPA, utilizing `triton.autotune` to sustain **16.2 TFLOP/s** (~40% of RTX 5050 peak).

Experience

Teaching Assistant, IE 332: Computing in Industrial Engineering
Purdue University

Aug 2023 – Dec 2025
West Lafayette, Indiana

- ★ Led lab sessions for 75 students on Machine Learning algorithms, data pipelines, and supervised/unsupervised learning.
- ★ Taught relational database design, ERDs, SQL, and backend web development (HTML/CSS/JS/PHP/MySQL).
- ★ Guided student projects integrating ML with ERP, CRM, and SCM systems.

Research Assistant
Purdue University | [GitHub](#)

Oct 2024 – Jan 2025
West Lafayette, Indiana

- ★ Validated clean-priority learning theory in neural networks by showing early preference for clean over noisy labels.
- ★ Analyzed test error curves and sample-wise gradient trends under varying label noise.
- ★ Ran experiments on MNIST and CIFAR-10 with CNNs, FCNs, and ResNets to study learning behavior.

Data Science Researcher
Cisco X Purdue Data Mine

Jan 2024 – May 2024
West Lafayette, Indiana

- ★ Built forecast selection & ensembling algorithms for 10k+ Cisco products, improving cost-weighted prediction accuracy by **3%**.
- ★ Clustered PLIDs using **Gaussian Mixture Models** to refine forecast segmentation.
- ★ Applied Bayesian MCMC, MLE, and advanced calibration methods to optimize ensemble weighting.